Víctor Cazalla
Mario Corrales
Laura Corredor

TFM proposal: Predictive analysis of suicide rates based on socioeconomic determinants and mental health prevalence.

1. Abstract

Mental health has emerged as a critical global challenge, deeply influenced by complex social and economic factors. While existing evidence underscores the correlation between socioeconomic determinants and psychological outcomes, the application of Machine Learning (ML) on multinational datasets offers a transformative opportunity to predict mental health indicators.

This project aims to develop a reproducible data pipeline utilizing open-access sources to perform cross-national comparisons and generate actionable insights for both public and private sectors. Specifically, the study seeks to provide strategic value for:

- Public Administrations: By informing health planning, preventive strategies, and the design of evidence-based public policies.
- Healthcare and Pharmaceutical Industries: By identifying emerging trends, forecasting future demands, and uncovering market opportunities across diverse geographic contexts.

2. Nature of the problem and justification

The primary objective of this research is to anticipate the evolution of suicide rates at a global scale by analysing their dependency on mental health indicators (disorder prevalence) and socioeconomic variables. This project aims to transform historical data into actionable insights for public administrations and the private sector, facilitating a shift from reactive management to proactive intervention.

From an analytical perspective, the problem will be addressed through a comprehensive workflow. Within the following, the main objective is the first one, where the rest will be considered in case the length of the project does not exceed the time constraints.

- Supervised Learning: Utilizing Regression models to predict suicide rates and Classification algorithms to segment countries or periods into different risk levels.
- Unsupervised Learning: Identifying patterns and clustering countries with similar public health and socioeconomic profiles.
- Temporal and Spatial Analysis: Examining historical evolution and employing geostatistical representation to detect regional disparities.
- Feature Importance Analysis: Identifying which factors (e.g., GDP, unemployment rates, depression prevalence) hold the greatest explanatory or predictive power.

3. Objectives

1. Multi-source Data Integration: Construct a unified, traceable, and reproducible dataset from official sources including Kaggle, the WHO, World Bank, and Eurostat.
2. Predictive and Descriptive Modelling: Develop robust models to estimate suicide rate trends and group regions according to their vulnerability.
3. Advanced Visualization: Implement interactive dashboards and geostatistical maps to facilitate the interpretation of results for strategic decision-making in both the public and private sectors.

4. Methodology and technology stack

A. Data Acquisition and Management

- Sources: Data extraction via official APIs and structured dataset downloads. Data sources will be further detailed in next section.
- Version Control: GitHub will be taken into consideration for code versioning and collaborative development.
- Storage: Google Drive will serve as the repository for raw and processed large-scale data files.
- Relational Logic: Python ecosystem will be used to handle complex joins between multinational datasets, ensuring data integrity and efficiency, as well as the posterior data cleaning, modelling and evaluation.

B. Analysis and Modelling

- EDA and Cleaning: Development in VS Code, utilizing Pandas and Numpy.
- Feature Engineering: Implementation of normalization, categorical encoding, and dimensionality reduction using scikit-learn.
- Proposed Models:
  - Clustering: Applying Principal Component Analysis (PCA) and K-Means to segment countries by socioeconomic profiles.
  - Forecasting/Regression: Implementing advanced algorithms such as XGBoost or Random Forest Regressor to predict suicide indicators.
- Evaluation Metrics:
  - Clustering: Silhouette Score and the Elbow Method (WCSS).
  - Regression: MAE (Mean Absolute Error), RMSE (Root Mean Squared Error), and R^2 Score.

C. Visualization and Reporting

- Power BI: Design of interactive dashboards for monitoring performance metrics and regional trends.
- QGIS (Geographic Information System): Generation of professional cartography and spatial analysis to visualize geographical distribution of risk factors.

Technologies and databases use is subject to change with the evaluation of the project and the discovery of new tools that could improve it.

Víctor Cazalla
Mario Corrales
Laura Corredor

5. Data bases proposed

Although data bases might contain a wide range of regions and countries, for the seek of simplification of this project, only Europeans countries will be considered. In addition, only the years when all databases merge will be considered. There will be no distinction between female and male population

- **Mental health**. Source: Kaggle.
  This will be the main database. It contains the yearly share of population affected by Schizophrenia, Depressive disorders, Anxiety, Bipolar disorders and eating disorders, all being age standardized.
  https://www.kaggle.com/datasets/imtkaggleteam/mental-health?select=4-+adult-population-covered-in-primary-data-on-the-prevalence-of-mental-illnesses.csv
- **European countries population**. Source: World Bank.
  https://data.worldbank.org/indicator/SP.POP.TOTL?locations=EU
- **GDP per capita per country and regions**. Source: World Bank.
  https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=EU
- **Health expenditure per country and regions**. Source: World Bank.
  https://data.worldbank.org/indicator/SH.XPD.CHEX.GD.ZS?locations=EU)
- **Causes of death standardised**. Source: Eurostat.
  Where the codes to be used will be E950-E959 or X60-84 (intentional self-harm - suicides), Mental and behavioural disorders, Substance abuse and mental health related codes.
  https://ec.europa.eu/eurostat/databrowser/view/hlth_cd_asdr2/default/table?lang=en
- **Number of psychiatrist beds**. Source: Eurostat.
  Fields to be used: psychiatric care.
  https://ec.europa.eu/eurostat/databrowser/view/hlth_rs_bds1__custom_20792933/default/table

As mentioned before, new databases might be included in case more fields of interest for this project are discovered.